



Project Packet Tracer

Computer Networks Lab

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Date: 12/05/2025



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Introduction

This report presents the configuration and implementation of a complex enterprise network simulation using Cisco Packet Tracer for the CS-3001



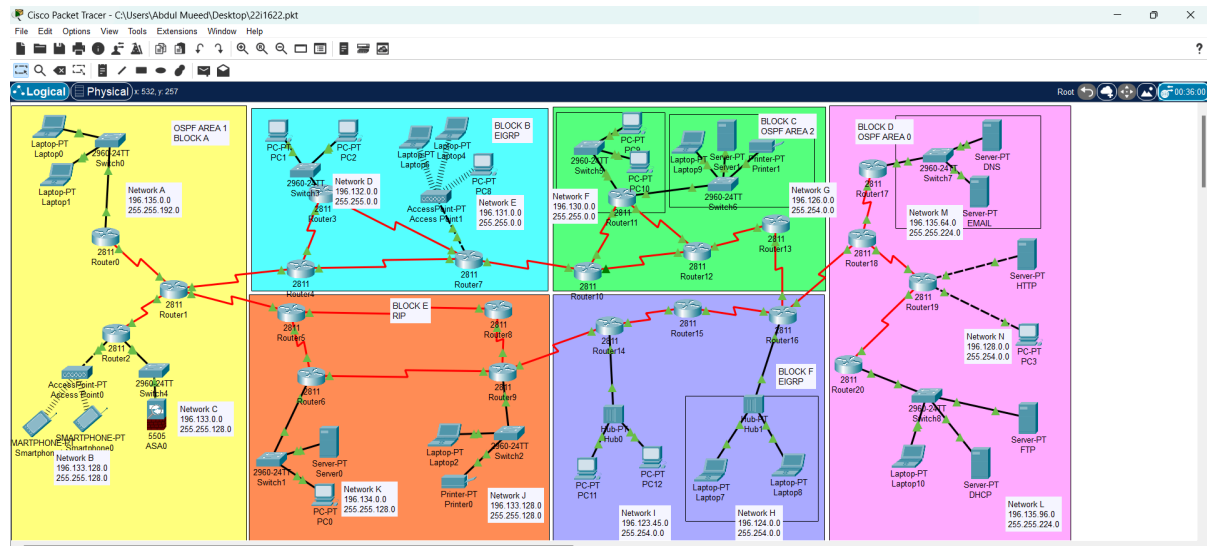
Computer Networks course. The goal of this project was to design, configure, and troubleshoot a multi-protocol network incorporating various networking technologies such as routing protocols (EIGRP, OSPF, RIP), Network Address Translation (NAT), Access Control Lists (ACLs), and several server applications (DHCP, SMTP, FTP, and Web). The project aims to emulate real-world networking scenarios where devices need to communicate across different networks with controlled access and efficient routing. Through this hands-on implementation, various network management aspects such as addressing, routing, security, and service configurations were explored and validated.

Topology creation

The network topology was designed according to the provided layout. All routers, switches, end devices (PCs, Laptops, Smartphones), and servers (FTP, Web, Email, DHCP) were placed and connected logically. Labeling was done to clearly identify each network block and interfaces.



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IP Addressing

- VLSM was used to efficiently allocate IPs based on the number of hosts required per subnet.
- All point-to-point links between routers were given /30 subnets.
- Internal LANs were subnetted based on required hosts.
- Each device (hosts and router interfaces) was assigned IPs within the appropriate subnets.
- Addressing plan ensures minimal wastage and future scalability.

The VLSM document was submitted before

#	Network	Usable IPs	Broadcast IP	Subnet Mask
1	10.0.0.0/30	10.0.0.1 – 10.0.0.2	10.0.0.3	255.255.255.252



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2	10.0.0.4/30	10.0.0.5 – 10.0.0.6	10.0.0.7	255.255.255.252
3	10.0.0.8/30	10.0.0.9 – 10.0.0.10	10.0.0.11	255.255.255.252
4	10.0.0.12/30	10.0.0.13 – 10.0.0.14	10.0.0.15	255.255.255.252
5	10.0.0.16/30	10.0.0.17 – 10.0.0.18	10.0.0.19	255.255.255.252
6	10.0.0.20/30	10.0.0.21 – 10.0.0.22	10.0.0.23	255.255.255.252
7	10.0.0.24/30	10.0.0.25 – 10.0.0.26	10.0.0.27	255.255.255.252
8	10.0.0.28/30	10.0.0.29 – 10.0.0.30	10.0.0.31	255.255.255.252
9	10.0.0.32/30	10.0.0.33 – 10.0.0.34	10.0.0.35	255.255.255.252
10	10.0.0.36/30	10.0.0.37 – 10.0.0.38	10.0.0.39	255.255.255.252
11	10.0.0.40/30	10.0.0.41 – 10.0.0.42	10.0.0.43	255.255.255.252
12	10.0.0.44/30	10.0.0.45 – 10.0.0.46	10.0.0.47	255.255.255.252
13	10.0.0.48/30	10.0.0.49 – 10.0.0.50	10.0.0.51	255.255.255.252
14	10.0.0.52/30	10.0.0.53 – 10.0.0.54	10.0.0.55	255.255.255.252
15	10.0.0.56/30	10.0.0.57 – 10.0.0.58	10.0.0.59	255.255.255.252
16	10.0.0.60/30	10.0.0.61 – 10.0.0.62	10.0.0.63	255.255.255.252
17	10.0.0.64/30	10.0.0.65 – 10.0.0.66	10.0.0.67	255.255.255.252
18	10.0.0.68/30	10.0.0.69 – 10.0.0.70	10.0.0.71	255.255.255.252
19	10.0.0.72/30	10.0.0.73 – 10.0.0.74	10.0.0.75	255.255.255.252
20	10.0.0.76/30	10.0.0.77 – 10.0.0.78	10.0.0.79	255.255.255.252
21	10.0.0.80/30	10.0.0.81 – 10.0.0.82	10.0.0.83	255.255.255.252
22	10.0.0.84/30	10.0.0.85 – 10.0.0.86	10.0.0.87	255.255.255.252
23	10.0.0.88/30	10.0.0.89 – 10.0.0.90	10.0.0.91	255.255.255.252
24	10.0.0.92/30	10.0.0.93 – 10.0.0.94	10.0.0.95	255.255.255.252

Routing Protocol Configuration

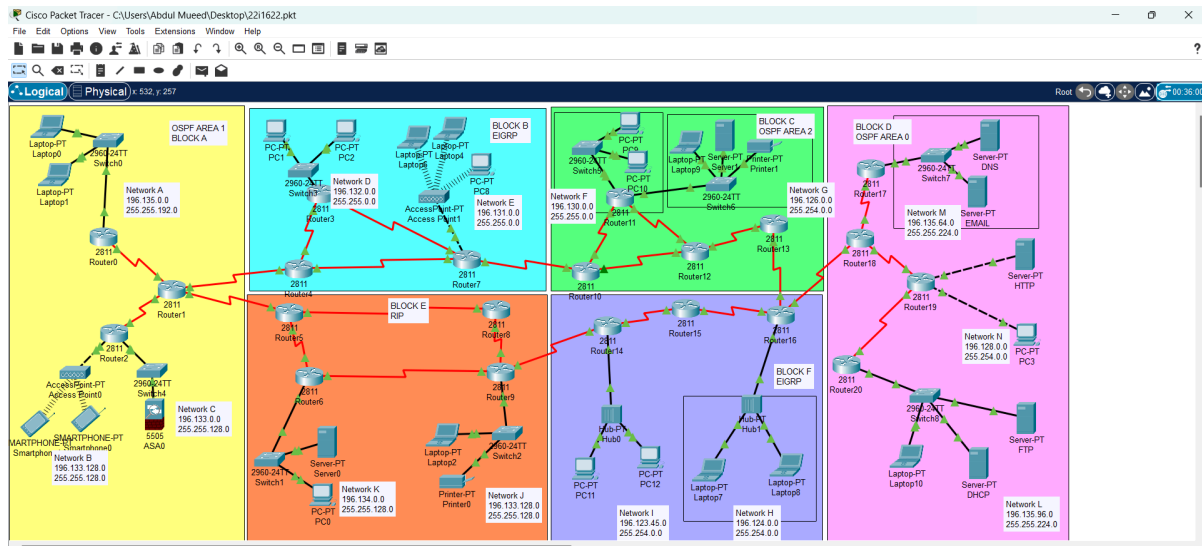
- EIGRP: Configured in the specified areas. Autonomous system number used: [e.g., 100].
- OSPF: Configured with Area 1 and Area 2. Verified using 'show ip ospf neighbor'.
- RIP: Configured using version 2 with only necessary interfaces advertised.
- Route Redistribution: Performed at border routers between routing domains. Route maps



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used.

- Routing Table: All networks are reachable from each other.

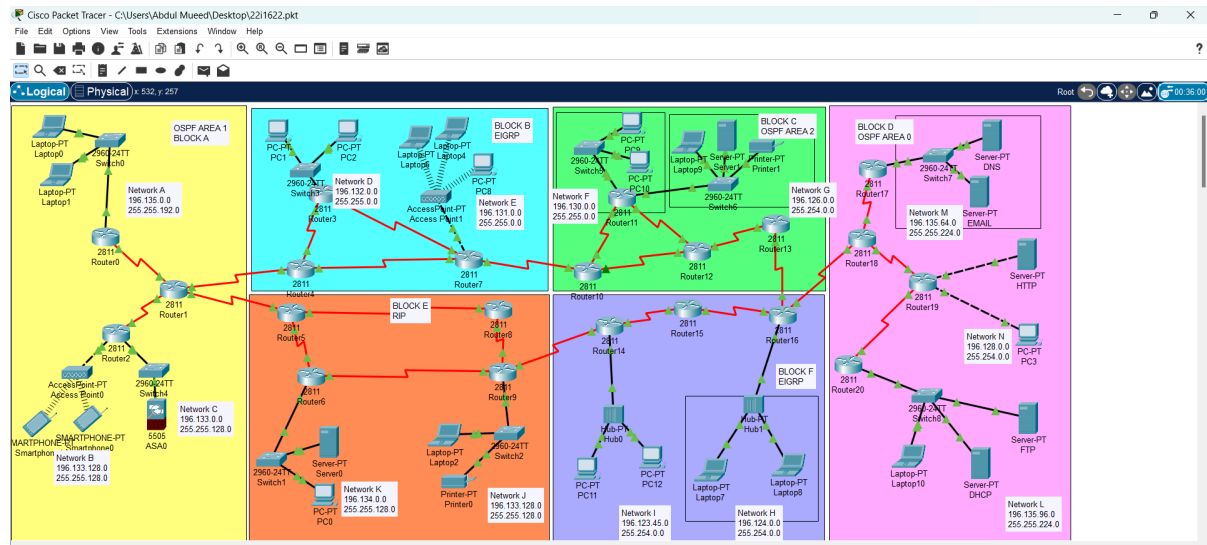


DHCP Server Configuration

- Centralized DHCP server placed in Block D.
- DHCP pools created for Networks in EIGRP, OSPF, and RIP zones.
- DHCP requests successfully received IPs.
- Routers configured as DHCP relay agents using 'ip helper-address'.



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NAT Configuration

- NAT configured on:
 - Router21 (Network K)
 - Router10 (Network F)
- Private IPs translated to public IPs.
- PAT (overload) used for dynamic address translation.
- Verified using 'show ip nat translations' and 'show ip nat statistics'.



HTTP Configuration

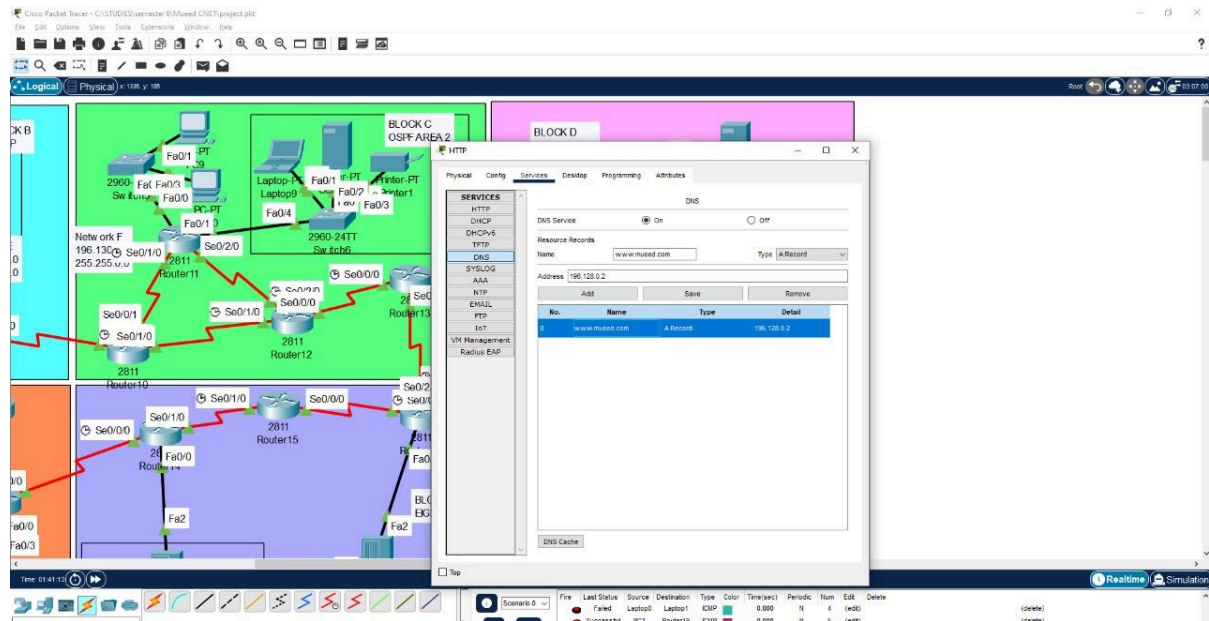
The screenshot displays the Cisco Packet Tracer interface. The main workspace shows a network topology with various devices including routers (2811, 2960, 2950), switches (2960, 2950), and end-user devices (Laptop, PC, Kiosk, Printer). The network is divided into colored blocks: BLOCK A (blue), BLOCK B (orange), BLOCK C (green), and BLOCK D (pink). A configuration window for a 2811 router is open, showing the 'Services' tab. The 'HTTP' service is configured as follows:

Service	Status
HTTP	On
CHCP	Off
DHCPv6	Off
TFTP	Off
DNS	Off
SYNLOG	Off
AAA	Off
NTP	Off
EMAIL	Off
FTP	Off
IoT	Off
VM Management	Off
Radius RADIUS	Off

The 'File Manager' section shows a list of files:

File Name	Edit	Delete
1. copyrights.html	(edit)	(delete)
2. ciscoptop177x111.jpg	(edit)	(delete)
3. helloworld.html	(edit)	(delete)
4. image.html	(edit)	(delete)
5. index.html	(edit)	(delete)

The bottom status bar indicates the simulation is running in 'Realtime' mode.

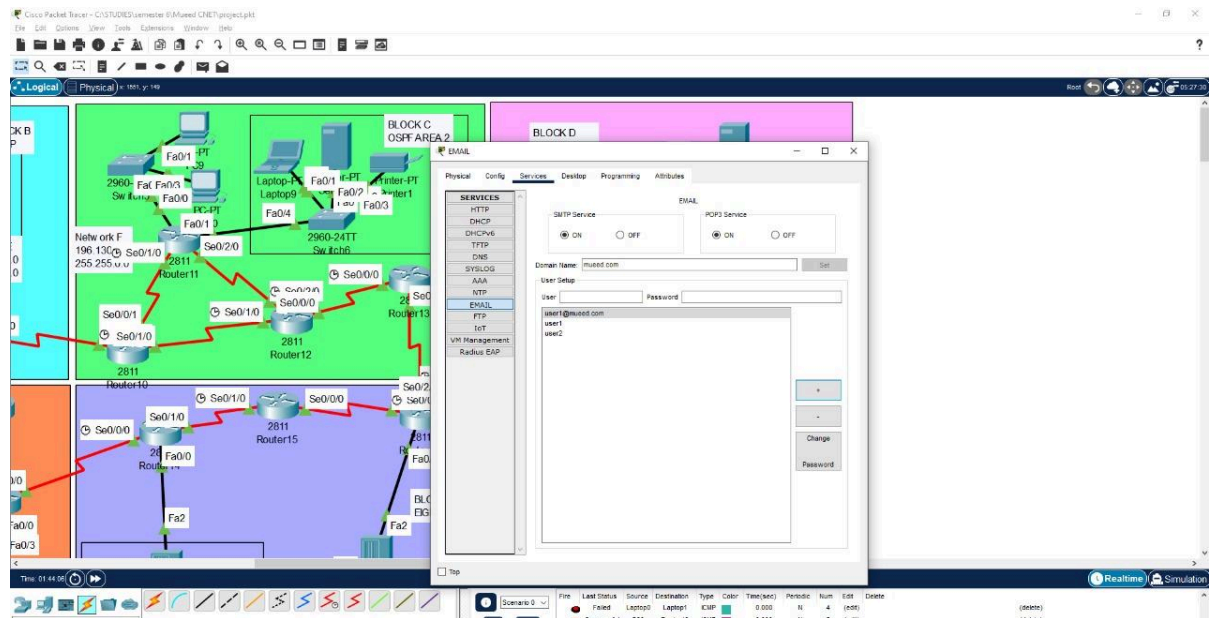


ACL Implementation

- ACL applied on the router connected to the Web Server.
- HTTP access denied for:
 - One PC from Network A
 - One Laptop from Network E
 - One Smartphone from Network B
 - All hosts from Network D
- Verified ACL behavior by testing blocked and allowed devices.



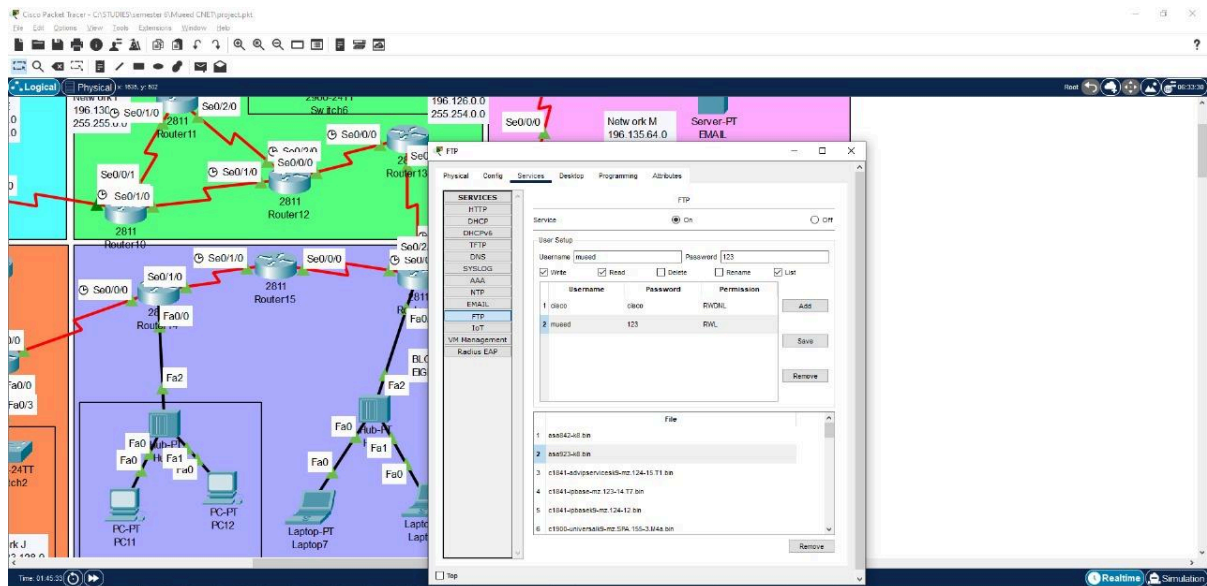
- SMTP and POP3 services enabled on Mail Server in Block D.
- Users in Network H and I configured with valid credentials.
- Email successfully sent and received using client email apps.



FTP Configuration

- FTP service enabled on server.
- Users in Network G created with upload permissions.

- Other networks denied access to FTP.
- Upload tested successfully from allowed hosts.



Conclusion

In conclusion, the successful configuration of a multi-protocol network in Cisco Packet Tracer demonstrates the ability to design and implement various network protocols and services for real-world application. The project covered essential topics such as IP addressing with VLSM, dynamic routing, NAT for private-to-public IP translation, and securing network services using ACLs.



Additionally, the deployment of FTP, email, and web services within the network showcases the integration of networking and server administration. By completing this project, a deeper understanding of network communication, troubleshooting, and security implementation was gained. The overall experience has provided valuable insights into real-world networking and the importance of efficient network management in maintaining secure and reliable communication.

References

1. Lab Manuals
2. ChatGPT
3. NAT:
<https://www.cisco.com/c/en/us/support/docs/ip/network-address-translation-nat/13772-12.html>
4. ACL:
<https://www.cisco.com/c/en/us/support/docs/security/ios-firewall/23602-confaccesslists.htmlw>
5. FTP: <https://youtu.be/Mk5WUsHOK0Y?si=QBWyHlcrk5wILFlv>